

CS101

Compiled vs Interpreted

2/19

Agenda

1. Compiled vs Interpreted (Lab6 Worksheet)
2. Demo costPerPerson.java
3. Work time
4. Windows OS Tips + Touch Typing

Announcements

1. Lab3 and HW2 due tomorrow 2/20 at midnight
2. Lab1 and Lab2 renamed and solution posted on GitHub
3. Poll → allow late labs for 50%?

Interpreter vs Compiler: What's the Difference?

Types of “code”

- Source Code
 - Human readable text files
 - .java .py .c
- Bytecode
 - Intermediate compiled code
 - .class
- Machine code
 - 1s and 0s (Hardware specific binary instructions)
 - .exe .app

COMPILED • INTERPRETED • HYBRID

CATEGORY	• COMPILED	• INTERPRETED	• HYBRID
HOW IT WORKS	Entire source → machine code.	Executed line-by-line at runtime.	Compiled to bytecode; run by Virtual Machine (VM).
SPEED	FAST	SLOWER	MEDIUM
PORTABILITY	Platform-specific binary. Recompile per OS.	Runs anywhere an interpreter exists.	Bytecode runs on any supported VM.
ERROR DETECTION	Caught at compile time.	Surfaces at runtime.	Compile-time
DEV SPEED	Slower – compile step each iteration.	Fast – edit and run instantly.	Moderate
LANGUAGES	C C++ Rust Go	Python JavaScript PHP	Java C# Kotlin Scala

Compiler and Interpreter: Compiled Language vs Interpreted Programming Languages

C

- Few packages → Little overhead
- Use cases:
 - Operating Systems (Windows, Linux, macOS)
 - Small devices (car engine controllers, smart light bulbs, etc)

Java

- Strict code organization
 - Pro: Safer, easier to collaborate
 - Cons: Harder to experiment
- Use cases
 - Netflix, Amazon, Android Apps

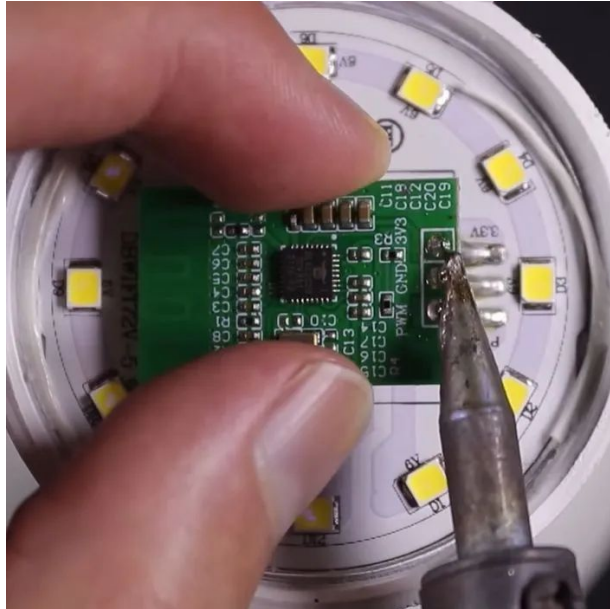
Python

- Use Cases
 - “fast enough” or “safe enough” applications
 - Quick prototyping
 - Designing AI models
 - Versatile
 - “2nd best option for everything”

Python Vs C++ Vs Java!

Example: Smart bulb CPU

[Video: Running minecraft on smart-bulb cpu](#)



Bouffalo Lab BL602 Chip

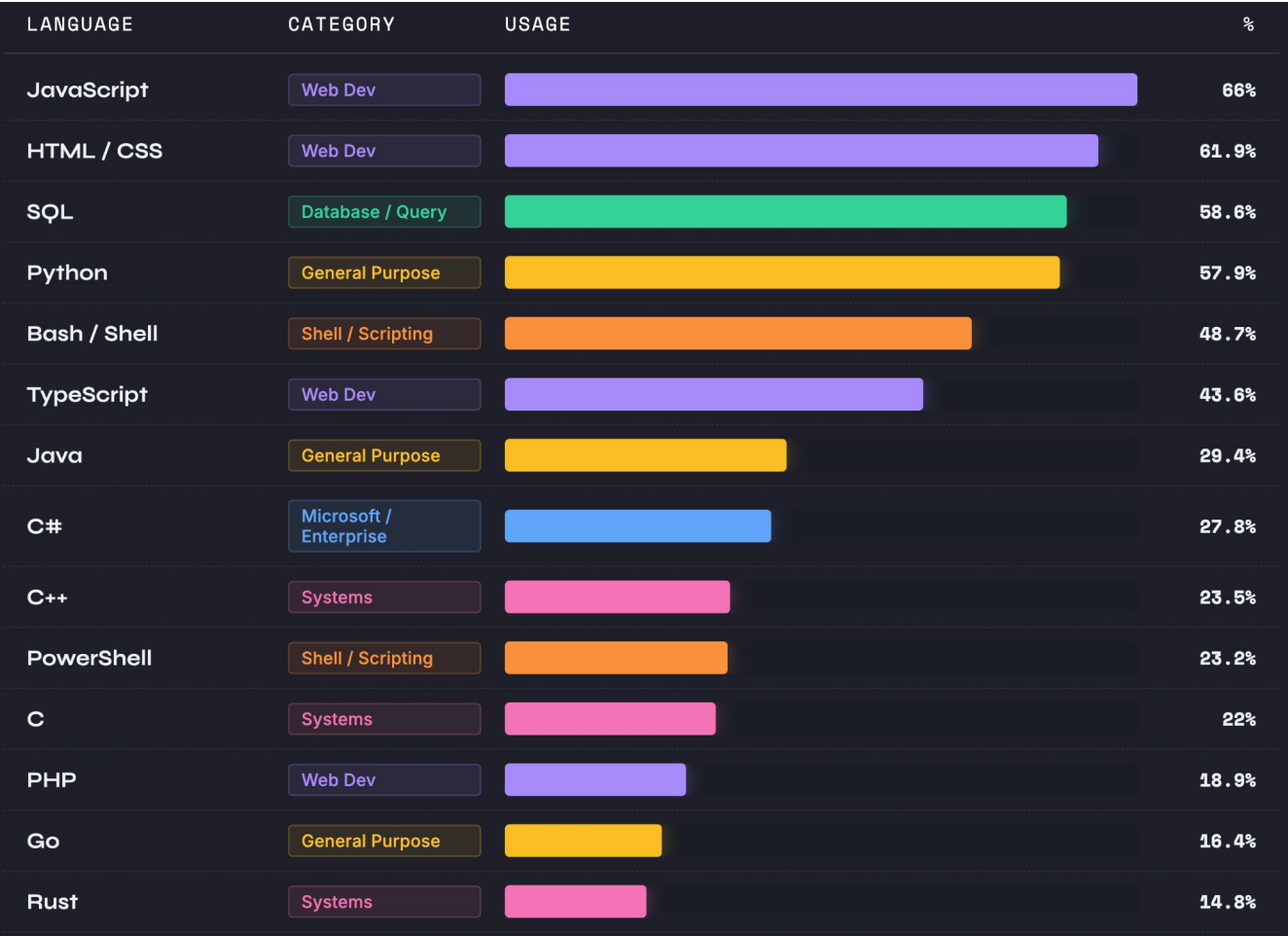
- CPU: 192 MHz
 - 1/10 of Dell Laptop 2.4GHz
- RAM: 276 KB
 - 1/1000 of Dell Laptop 8GB
- Storage: 128 KB
 - 8 Seconds of a song
- Wifi and Bluetooth capable

HARDWARE REQUIREMENTS FOR "HELLO, WORLD" APP

REQUIREMENT	COMPILED C	COMPILED JAVA	PYTHON SCRIPT
File Type	Binary	.class (bytecode)	.py
Software Required	None	JVM	Python Interpreter
CPU Speed	~1 MHz	~200 MHz	~200 MHz
RAM	~1 KB	~64–128 MB	~25–50 MB
Storage	Size of binary (KB range)	~150–200 MB	~30–100 MB

Global Usage (Stack Overflow Survey)

<https://survey.stackoverflow.co/2025/technology/>



Demo: costPerPerson.java

Lab4/Quiz4 Review

Lab5 Work time